

Kidney Ultrasound Pathology Classification

A dataset-grounded six-week proposal | Four-person team | For instructor review, not an approved clinical workflow

Research objective. Evaluate whether kidney-specific representation learning and transfer learning improve the detection of pathological findings in B-mode kidney ultrasound images, relative to standard image-pretrained classification baselines. The initial outcome is healthy vs pathological; anatomical segmentation is investigated as a supporting task rather than assumed to be required.

1. Project motivation and research scope

Ultrasound is inexpensive and portable, but interpretation requires expertise. A research classifier could eventually support triage in primary care and reduce unnecessary referrals; this project evaluates technical feasibility rather than claiming clinical readiness. [1, 4]

Research questions

- RQ1. Can a model reliably distinguish healthy from pathological kidney ultrasound images on held-out clinical cases?
- RQ2. Does kidney-domain pretraining improve performance compared with training from scratch or ImageNet initialization?
- RQ3. Does kidney-focused segmentation or region processing add predictive value beyond full-image classification?
- RQ4. How robust are results to class imbalance, ultrasound quality, image perspective and small localized lesions?

2. Available reference datasets and verified properties

Resource	Verified description	Role / important limitation
URI-CADS dataset (2024) [4]	1,985 B-mode images: 450 healthy; 1,535 pathological. Includes kidney masks, global pathology indicators and lesion bounding boxes.	Primary candidate for diagnostic classification; confirm that the actual archive, annotations and permissions are in the local project folder.
Open Kidney Ultrasound [2, 3]	514 unique PNG images plus 20 repeated copies (534 total). Quality/view labels and polygons for capsule, cortex, medulla and central echo complex.	Candidate for anatomy segmentation and feature pretraining. The documented annotations are not a validated binary diagnosis target.
~5,000-image resource [1]	A second ~5,000-image kidney dataset / existing pretraining was discussed with the instructor.	Unverified in supplied dataset documents. Identify source, checkpoint, architecture and permitted use before including as an experiment.

Data integrity gate: Verify image count, annotation files, patient/case IDs, duplicates, file formats, licensing and potential cross-dataset overlap before model training. Never label an image “healthy” merely because no segmentation polygon was

drawn. [2, 3, 4]

3. Research design and staged experiments

Stage	Method	Question answered / deliverable
E0: Data audit	Inspect dataset structure, diagnoses, pathology taxonomy and quality labels; perform EDA and identify case-safe splits.	Dataset inventory, label dictionary, exclusion rules and defensible test protocol.
E1: Baselines	Simple CNN from scratch; ImageNet-pretrained ResNet-18/50; optionally EfficientNet-B0.	Reproducible healthy/pathological performance reference using identical splits.
E2: Transfer learning	If available, fine-tune verified kidney-specific pretrained weights; alternatively train a compatible anatomy-segmentation encoder and transfer features.	Controlled comparison of initializations and transfer strategies.
E3: Anatomy-aware models	Compare original full image vs kidney crop/mask-assisted input; investigate multi-task model only if time, annotations and compute permit.	Ablation isolating whether explicit anatomical information improves diagnosis.

Reference implementation. URI-CADS uses a ResNet-50 backbone with segmentation, image-level classification and region-level detection branches, based in part on Mask R-CNN and Faster R-CNN. It is a literature benchmark, not a requirement to reproduce the complete system. Its published binary ROC-AUC was 87.41%; this is not an expected result for the team. [4]

4. Practical training and evaluation protocol

- Splits and leakage. Prefer subject/case-level stratification with a held-out test set or grouped cross-validation where identifiers are available. If identifiers are missing, document this limitation and use the strongest defensible duplicate and overlap checks. Do not tune thresholds on the test set.
- Preprocessing. De-identification check; inspect machine overlays; standardize pixel range and image size; retain appropriate clinical context. Fit all data-driven preprocessing on training data only.
- Augmentation. Apply realistic small rotations, translation, zoom and intensity/gamma adjustments only to training examples; validate flips with clinical experts. Follow source-image and quality-label constraints. [3, 4]
- Model setup. PyTorch; binary cross-entropy or BCE-with-logits; AdamW as proposed training choices. Select class weighting only after reviewing class prevalence. Tune models within training/validation folds.
- Primary evaluation. ROC-AUC, sensitivity, specificity and specificity at a clinically motivated sensitivity threshold; report precision, F1, PR-AUC, confusion matrices and uncertainty intervals where feasible. Include per-pathology and quality-stratified error analysis. [4]
- Segmentation measures. For a segmentation subtask, Dice/IoU are appropriate; do not confuse anatomical segmentation accuracy with diagnostic classification accuracy.

5. Team allocation (four members)

Lead	Responsibilities	Deliverables
Member 1 - Data & protocol	Dataset inventory, labels, quality distributions, patient/case-level splitting, leakage checks.	Data audit report and versioned manifests.
Member 2 - Literature & transfer	Study URI-CADS/Open Kidney material; verify available checkpoints; establish the domain-transfer experiment.	Literature matrix and transfer setup.
Member 3 - Modeling	Implement simple CNN and image-pretrained baseline(s); track hyperparameters and compute budget.	Training scripts, experiment logs and comparisons.
Member 4 - Evaluation & reporting	Shared metrics, calibration/error analysis, explainability and optional segmentation ablation; integrate report.	Evaluation notebook, figures and slides.

All members jointly approve the label protocol, data splitting and final results; each experiment should have one code reviewer. Dataset files and patient information must be handled under the director's access conditions.

6. Six-week provisional schedule

Period	Milestone	Evidence required
11-16 Oct	Confirm scope, inspect local archives and labels, resolve dataset identity and compute access.	Dataset inventory + question list for instructor (16 Oct, time unconfirmed).
17-23 Oct	Complete reading, data protocol, duplicate checks, splits, preprocessing pipeline.	Methodology note, reproducible loader and quality summary.
24-30 Oct	Run baseline CNN and ImageNet transfer learning.	First valid performance table and error examples.
31 Oct-6 Nov	Test verified kidney-domain pretraining, if viable.	Same-split comparison, training logs and ablation.
7-13 Nov	Assess reliability and explore segmentation-assisted classification.	ROC/PR curves, subgroup checks, optional mask experiments.
14-20 Nov	Repeat key runs and synthesize final findings.	Final report, limitations, presentation and repository documentation.

Dates are planning targets and should be revised after access/label verification; the instructor follow-up is provisionally 16 October 2026. [1]

7. Open decisions for the next meeting

- [] Which local folder corresponds to each documented dataset, and are all images and annotations present?
- [] What diagnosis label mapping is clinically acceptable (including images with multiple findings)?
- [] Is the previously mentioned ~5,000-image model available as actual pretrained weights? What is its architecture?
- [] Can we identify independent patients/cases and rule out cross-dataset overlap?
- [] Is the expectation classification only, or an explicit segmentation/detection experiment?
- [] What GPU access, privacy restrictions, signed paperwork and deliverable deadlines apply?

8. Risks, limits and decision gates

Risk	Response
No usable diagnostic labels	Do not train a pathology classifier on anatomy/quality labels. Request the clinical label file or redefine scope with instructor approval.
Unverified pretrained checkpoint	Proceed with ImageNet baselines; postpone domain-specific comparisons until the checkpoint and data provenance are confirmed.
Limited or skewed clinical cases	Use grouped folds, class-aware assessment, uncertainty intervals and avoid inflated accuracy from duplicated images.
Segmentation too costly	Keep a valid classification deliverable; mark anatomy-aware work as optional and prioritize interpretable comparisons.
Possible data restrictions	Keep original medical data out of public GitHub until permission is explicitly confirmed; publish code, documentation and approved aggregates only.

9. Source documents and traceability

[1] Instructor meeting report / project description (provided by project team; planned meeting 16 October 2026).

[2] Singla et al., The Open Kidney Ultrasound Data Set, dataset datasheet, Appendix A (provided PDF).

[3] Kidney Ultrasound Tracing Annotation Instructions, version 1.3, 3 May 2021 (provided DOCX).

[4] Molina-Moreno et al., URI-CADS: A Fully Automated Computer-Aided Diagnosis System for Ultrasound Renal Imaging, Journal of Imaging Informatics in Medicine (2024), DOI: 10.1007/s10278-024-01055-4.

Document status: Proposed research plan. Counts and features above refer to the supplied documents; the content of the actual local archives and availability of model checkpoints remain to be independently confirmed.